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VOLUNTEER MANAGEMENT SYSTEM FOR COLLEGE EVENTS

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

VISION AND MISSION OF THE INSTITUTION

VISION

To emerge as a leader among the top institutions in the field of technical education.

MISSION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges.
- Create a diverse, fully-engaged learner-centric campus environment to provide quality education to the students.
- Maintain mutually beneficial partnerships with our alumni, industry and professional associations.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VISION

To empower the Electronics and Communication Engineering students with emerging technologies, professionalism, innovative research and social responsibility.

MISSION

M1: Attain the academic excellence through innovative teaching learning process, research areas & laboratories and Consultancy projects.

M2: Inculcate the students in problem solving and lifelong learning ability.

M3: Provide entrepreneurial skills and leadership qualities.

M4: Render the technical knowledge and skills of faculty members.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Core Competence: Graduates will have a successful career in academia or industry associated with Electronics and Communication Engineering.

PEO2: Professionalism: Graduates will provide feasible solutions for the challenging problems through comprehensive research and innovation in the allied areas of Electronics and Communication Engineering.

PEO3: Lifelong Learning: Graduates will contribute to the social needs through lifelong learning, practicing professional ethics and leadership quality.

PROGRAM OUTCOMES(POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health, safety, cultural, societal, and environmental considerations.

PO4: Conduct Investigations of Complex Problems: Use research-based knowledge and research methods, including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities with an understanding of limitations.

PO6: The Engineer and The World (Includes Sustainability): Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate knowledge of and need for sustainable development.

PO7: Ethics (Expanded): Apply ethical principles, commit to professional ethics and responsibilities, and include diversity and inclusive behavior in engineering practices.

PO8: Individual and Collaborative Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Communication (Inclusive): Communicate effectively on complex engineering activities with the engineering community and society at large, including the ability to comprehend and write effective reports, design documentation, and make effective presentations.

PO10: Project Management and Finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects.

PO11: Life-Long Learning (Adaptability): Recognize the need for and prepare to engage in independent and life-long learning, including adaptability to new and emerging technologies and critical thinking in response to technological changes.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Applying knowledge in various areas, like Electronics, Communications, Signal processing, VLSI, Embedded systems etc., in the design and implementation of Engineering application.

PSO2: Able to solve complex problems in Electronics and Communication Engineering with analytical and managerial skills either independently or in team using latest hardware and software tools to fulfil the industrial expectations.

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We gratefully remember our beloved **Founder Chairman, (Late) Thiru. M. Kumarasamy**, whose vision and legacy laid the foundation for our education and inspired us to successfully complete this project.

We extend our sincere thanks to **Dr. K. Ramakrishnan, Chairman**, and **Mr. K. R. Charun Kumar, Joint Secretary**, for providing excellent infrastructure and continuous support throughout our academic journey.

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ABSTRACT

The Volunteer Management System is a web-based application developed to efficiently manage volunteer events and participant activities. It provides a centralized platform for administrators to create and manage events, handle student applications, and monitor volunteer participation in an organized manner. Administrators can select volunteers based on criteria such as priority, skills, and attendance, ensuring effective allocation of resources for each event. Students can easily browse available events, view details, and apply according to their interests, improving engagement and participation. The system also integrates a microcontroller-based attendance mechanism, where student entry and exit are recorded automatically. The attendance status is displayed on an LCD screen along with the student's registration number, ensuring accuracy and reducing manual errors. All data is securely stored in a database, allowing quick access and efficient management. Overall, the system reduces manual effort, improves transparency, and provides a reliable solution for managing volunteer activities in institutions.

Abstract (Key words)	POs Mapping
Volunteer Management System, Web-Based Application, Event Management, Automated Attendance, Microcontroller, RFID/Scanning, Database Management, Real-Time Monitoring, Smart System.	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PSO1, PSO2

SDG Goal		Remarks
SDG 4	This project supports Sustainable Development Goal by promoting digital efficiency, reducing manual workload, and enabling smarter management of volunteer activities.	Improves transparency and accuracy through automated attendance and efficient event management.

Project Component	Relevant IEEE Standards
Web-Based Application	IEEE 830
Database Management	IEEE 1012
Microcontroller Integration	IEEE 1149
Reliability & Performance	IEEE 1633
Documentation & testing	IEEE 830, IEEE 829

TABLE OF CONTENTS

CHAPTER No.	CONTENTS	PAGE No.
	INSTITUTION VISION AND MISSION	iii
	DEPARTMENT VISION AND MISSION	iii
	DEPARTMENT PEOs, POs AND PSOs	iv
	ABSTRACT	vii
	LIST OF TABLES	xii
	LIST OF FIGURES	xiii
	LIST OF ABBREVIATIONS	xiv
1	**INTRODUCTION**	**1**
	1.1 Overview of Volunteer Management Systems	1
	1.2 Need for Automation in Event and Attendance Management	2
	1.2.1 Limitations of Manual Event Management	3
	1.2.2 Impact of Inefficient Attendance Tracking	3
2	**LITERATURE SURVEY**	**4**
	2.1 Web-Based Event Management Systems	4
	2.2 Automated Attendance Systems using RFID	4
	2.3 Smart Campus Management Solutions	4
	2.4 Database-Driven Student Management Systems	5

	2.5 Role of IoT in Attendance Monitoring	5
	2.6 Wireless Communication in Smart Systems	5
	2.7 Real-Time Monitoring and Tracking Systems	5
	2.8 User-Friendly Interface Design in Web Applications	6
	2.9 Integration of Hardware with Web Applications	6
	2.10 Modern Approaches to Event and Volunteer Management	6
3	**EXISTING SYSTEM**	**7**
	3.1 Manual Event Registration Methods	7
	3.2 Paper-Based Attendance Tracking	7
	3.3 Basic Digital Record Systems	8
	3.4 Lack of Centralized Management	8
	3.5 Delayed Data Processing	8
	3.6 Limited Transparency in Volunteer Selection	9
	3.7 Absence of Real-Time Monitoring	9
	3.8 Inefficient Resource Allocation`	10
4	**PROBLEM STATEMENT**	11
5	**PROPOSED SYSTEM**	12

6	**SOFTWARE AND HARDWARE DESCRIPTION**	18
7	**RESULTS AND DISUSSION**	22
8	**CONCLUSION AND FUTURE WORK**	25
	APPENDICES	27
	REFERENCES	42

LIST OF TABLES

TABLE No.	TITLE	PAGE No.
5.1	Proposed System Components	15
7.1	Attendance Update Status	24
7.2	Experimental Results	24

LIST OF FIGURES

FIGURE No.	TITLE	PAGE No.
5.1	Flow Diagram	14
5.2	Block Diagram	15
5.3	Front-End (User Interface)	16
5.4	Back-End (Server Side)	16
5.5	Hardware Setup	17
5.6	Attendance Marking	17
6.1	Circuit Diagram	21

LIST OF ABBREVIATIONS

ACRONYM		ABBREVIATION
RFID	-	Radio Frequency Identification
LCD	-	Liquid Crystal Display
MCU	-	Microcontroller Unit
HTML	-	HyperText Markup Language
CSS	-	Cascading Style Sheets
PHP	-	Hypertext Preprocessor
Wi-Fi	-	Wireless Fidelity

CHAPTER 1

INTRODUCTION

Volunteer Management Systems are widely used in educational institutions and organizations to efficiently manage events, volunteers, and participation activities. These systems provide a centralized platform to handle event creation, student registration, application processing, and attendance tracking. In modern institutions, managing volunteer activities manually becomes difficult due to the increasing number of students and events.

A web-based Volunteer Management System helps in organizing all activities in a structured manner. It reduces manual effort, improves transparency, and ensures accurate data handling. The integration of hardware components like microcontrollers and RFID-based attendance systems further enhances automation and reliability.

1.1 Overview of Volunteer Management Systems

A Volunteer Management System consists of multiple modules such as admin panel, student interface, event management, application tracking, and attendance monitoring. The admin can create and manage events, select participants, and monitor overall activity, while students can view available events and apply based on their interests.

The system uses a database to store all information such as student details, event data, and attendance records. The integration of microcontroller-based scanning systems allows automatic attendance marking when students enter and exit the event location.

Institutions prefer such systems because:

- They reduce manual paperwork and administrative workload
- They reduce manual paperwork and administrative workload
- Provide faster and more accurate data processing
- Improve transparency in volunteer selection
- Enable real-time monitoring of activities

The efficiency of the system depends on proper database management, smooth user interface, and accurate attendance tracking mechanisms.

1.2 Need for Automation in Event and Attendance Management

In traditional systems, event management and attendance tracking are performed manually, which leads to errors, delays, and inefficiency. As the number of students and events increases, managing data becomes complex and time-consuming.

Automation is required to simplify these processes and ensure accuracy. A smart system can handle event creation, application processing, and attendance tracking automatically without human intervention.

In addition, real-time attendance monitoring using RFID or scanning systems helps in improving accuracy and reducing manipulation.

Thus, institutions require an automated system that can efficiently manage events and track attendance in a reliable and scalable manner.

1.2.1 Limitations of Manual Event Management

- Event registration and tracking are time-consuming
- Data is often recorded on paper, leading to errors and loss
- Difficult to manage large number of student applications.
- No proper system for filtering or selecting volunteers
- No centralized system to store and access data.

1.2.2 Impact of Inefficient Attendance Tracking

The consequences of poor attendance management include:

- Incorrect attendance records due to manual entry errors
- Possibility of proxy attendance or manipulation
- Lack of real-time monitoring of participants
- Time delay in updating attendance records
- Impact of Inefficient Attendance Tracking

These challenges highlight the need for an automated attendance system integrated with event management, ensuring accuracy, efficiency, and transparency.

CHAPTER 2

LITERATURE REVIEW

2.1 Web-Based Event Management Systems

Author: R. Kumar and S. Gupta (2022) [1]

This paper presents a web-based event management system designed to simplify event creation, registration, and participant tracking. The system uses a centralized database to store event and user information. The study highlights improved efficiency, reduced manual errors, and better coordination between administrators and participants.

2.2 Automated Attendance Systems using RFID

Author: A. Sharma and P. Verma (2023) [2]

The authors propose an RFID-based attendance system where students scan their ID cards to mark attendance automatically. The system reduces manual errors and prevents proxy attendance. Experimental results show high accuracy and real-time data recording, making it suitable for educational institutions.

2.3 Smart Campus Management Solutions

Author: L. Wang and H. Liu (2021) [3]

This study focuses on integrating various campus activities such as event management, attendance, and communication into a single smart platform. The system uses IoT and web technologies to enable automation and real-time monitoring. Results indicate improved operational efficiency and better student engagement.

2.4 Database-Driven Student Management Systems

Author: M. Singh and K. Patel (2022) [4]

The paper discusses the role of database systems in managing student information efficiently. It highlights the importance of structured data storage, quick retrieval, and secure access. The implementation shows improved data accuracy and faster processing compared to manual systems.

2.5 Role of IoT in Attendance Monitoring

Author: S. Reddy and V. Kumar (2023) [5]

This research explores the use of IoT devices such as sensors and microcontrollers for automated attendance tracking. The system enables real-time monitoring and remote access to attendance data. The study demonstrates increased reliability and reduced human intervention.

2.6 Wireless Communication in Smart Systems

Author: J. Chen and Y. Lin (2021) [6]

The authors analyze the use of wireless communication technologies in smart management systems. The paper evaluates performance factors such as data transmission speed, reliability, and coverage. Results show that wireless systems enhance flexibility and ease of deployment.

2.7 Real-Time Monitoring and Tracking Systems

Author: K. Sharma and R. Das (2022) [7]

This paper presents a real-time monitoring system that tracks user activities and updates data instantly. The system ensures transparency and quick decision-making. Experimental analysis shows improved efficiency in managing large-scale operations.

2.8 User Interface Design in Web Applications

Author: N. Gupta and P. Mehta (2021) [8]

The study focuses on designing user-friendly interfaces for web applications. It emphasizes simplicity, responsiveness, and ease of navigation. Results indicate that a well-designed interface improves user experience and increases system usability.

2.9 Integration of Hardware with Web Applications

Author: T. Raj and S. Iyer (2023) [9]

This paper discusses the integration of hardware components such as microcontrollers and sensors with web-based systems. The system enables automatic data collection and real-time updates. The study highlights improved automation and accuracy in monitoring applications.

2.10 Modern Approaches to Event and Volunteer Management

Author: D. Brown and A. Smith (2020) [10]

This review paper analyzes modern techniques used in event and volunteer management systems. It includes cloud-based platforms, mobile applications, and automated selection processes. The study shows increased efficiency, scalability, and better resource utilization.

CHAPTER 3
EXISTING SYSTEM

3.1 Manual Event Registration Methods

In many educational institutions, event registration is still carried out using manual methods such as paper forms or verbal communication. Students are required to fill out physical forms or write their names in registers to participate in events. This approach is time-consuming and prone to errors such as missing or incorrect data entries.

Additionally, managing a large number of applications becomes difficult, especially when multiple events are conducted simultaneously. Sorting and filtering student applications manually is inefficient and increases the workload for administrators. There is also no proper record maintenance, which makes it hard to track participation history. As a result, the overall process becomes slow, unorganized, and unreliable.

3.2 Paper-Based Attendance Tracking

Attendance in many institutions is recorded manually using paper registers or sheets. Volunteers sign their names during entry and exit, and staff members later verify the records. This method is highly prone to errors such as duplicate entries, missed records, and illegible handwriting.

Moreover, there is a high possibility of proxy attendance, where one student signs on behalf of another. The process is time-consuming and lacks accuracy. Maintaining and storing physical records for long periods is also difficult. Overall, paper-based attendance systems are inefficient and unsuitable for modern large-scale event management.

3.3 Basic Digital Record Systems

Some institutions use simple digital tools such as spreadsheets or standalone software to manage event and student data. While these systems reduce paperwork, they still lack automation and integration. Data entry is often done manually, which can lead to errors and inconsistencies.

These systems do not support real-time updates or multi-user access, making it difficult for administrators to coordinate efficiently. Additionally, there is no proper linkage between event registration and attendance tracking. As a result, the system remains partially automated and does not fully address management challenges.

3.4 Lack of Centralized Management

In traditional systems, event management, student data, and attendance records are maintained separately. There is no centralized platform to integrate all these functions. This leads to data duplication, inconsistencies, and difficulty in accessing information.

Administrators must switch between multiple records or systems to manage a single event, which increases complexity and reduces efficiency. The absence of a unified system also makes reporting and analysis difficult. Hence, lack of centralization leads to poor coordination and ineffective management.

3.5 Delayed Data Processing

Manual and semi-digital systems often involve delays in data processing. Event applications, approvals, and attendance records are updated only after manual verification. This delay affects decision-making and reduces the responsiveness of the system.

For example, administrators cannot instantly know how many students have applied or attended an event. This lack of real-time information leads to inefficiencies in planning and execution. Delayed processing also increases the chances of errors and miscommunication.

3.6 Limited Transparency in Volunteer Selection

In many existing systems, the process of selecting volunteers is not transparent. Selection is often done manually based on incomplete or unorganized data. Students may not clearly understand the selection criteria, leading to dissatisfaction and reduced trust.

There is no proper mechanism to track or justify why certain students are selected over others. This lack of transparency can lead to bias and unfair practices. A structured and automated system is required to ensure fairness and clarity in the selection process.

3.7 Absence of Real-Time Monitoring

Traditional systems do not provide real-time monitoring of events and participant activities. Attendance and participation details are updated only after the event is completed. This makes it difficult for administrators to track ongoing activities.

Without real-time monitoring, it is not possible to verify whether students are present at the event location or not. This reduces accountability and increases the chances of incorrect records. Real-time tracking is essential for efficient event management.

3.8 Inefficient Resource Allocation

Due to lack of proper data and automation, resource allocation in events is often inefficient. Administrators may select more or fewer volunteers than required due to inaccurate information. This leads to poor utilization of manpower.

In some cases, important tasks may remain unattended due to lack of proper planning. Without a structured system, it is difficult to match student skills and availability with event requirements. This results in reduced productivity and ineffective event execution.

CHAPTER 4

PROBLEM STATEMENT

In educational institutions, managing volunteer events and student participation manually is a challenging and inefficient process. As the number of students and events increases, maintaining records, tracking applications, and managing attendance becomes complex and time-consuming. This often leads to data errors, mismanagement, and reduced overall efficiency.

Traditional methods such as paper-based registration and manual attendance tracking are slow, unreliable, and prone to mistakes. These systems do not provide real-time updates, making it difficult for administrators to monitor ongoing events or verify student participation accurately. Another major issue is the lack of transparency in volunteer selection and participation tracking.Manual systems also fail to prevent issues such as proxy attendance and incorrect record maintenance, reducing the reliability of the overall process.

To overcome these limitations, there is a need for an automated and centralized system that can efficiently manage events, track student participation, and monitor attendance in real time. The system should reduce manual effort, improve accuracy, and ensure transparency in operations.

This project aims to develop a web-based Volunteer Management System integrated with a microcontroller-based attendance mechanism. The proposed system is designed to be user-friendly, reliable, and efficient, enabling institutions to manage volunteer activities effectively while ensuring accurate data handling and improved productivity.

CHAPTER 5

PROPOSED SYSTEM

The proposed system is designed to efficiently manage volunteer events and student participation using a web-based platform integrated with an automated attendance monitoring system. The system addresses the limitations of manual methods by providing a centralized, real-time, and user-friendly solution for event creation, application handling, and attendance tracking.

In this system, administrators can create events, define details such as date, time, venue, and number of volunteers required. Students can view available events and apply based on their interests through a web interface. All the data is stored in a centralized database, ensuring easy access and management.

The attendance system is integrated with a microcontroller-based mechanism using RFID or scanning technology. When a student enters or exits the event location, their attendance is automatically recorded. This data is updated in real time and can be viewed by the administrator instantly. The system also displays attendance status on an LCD screen along with the student's registration number, ensuring transparency and accuracy.

Key Features

- **High-accuracy pressure detection:** Uses calibrated sensors to measure real-time pressure variations.
- **24/7 automated monitoring:** Eliminates the need for manual inspection.
- **Early fault detection:** Reduces maintenance downtime and equipment wear.
- **Energy efficiency:** Helps industries minimize compressed air wastage.
- **Scalable system:** Can be upgraded with wireless data transmission or IoT

connectivity.

- **User-friendly design:** Easy installation and simple configuration.

Working Principle

The proposed system works by continuously monitoring the pressure inside the pneumatic line using a pressure sensor. In a normal industrial environment, compressed air flows through pipes and valves at a constant pressure. When a leak occurs, the pressure in the pipeline begins to drop immediately. This change in pressure is sensed by the pressure sensor and sent as an electrical signal to the microcontroller.

The microcontroller (Arduino/ESP32) plays a key role in processing and interpreting the sensor readings. It compares the current pressure value with predefined threshold levels. For example, a value above 40 indicates high pressure, a value between 25–40 indicates normal pressure, and a value below 25 indicates a leak. When the sensor reading drops below the leak threshold, the microcontroller confirms the abnormality by checking the reading for a short duration to avoid false triggers caused by momentary pressure fluctuations.

Once the system confirms a leak, the microcontroller instantly activates multiple alert mechanisms. The LCD display shows “Leak Detected” or “High Leakage,” providing real-time information to the operator standing near the machine. At the same time, a buzzer and LED are turned on to give an audible and visual warning so that nearby workers can respond immediately.

If GSM communication is included, the microcontroller also sends an SMS alert to the maintenance team. This ensures that even if no operators are near the machine, the maintenance staff will be notified instantly. The message contains details such

as the location of the leak and the severity level, helping the team take corrective action quickly.

Through this continuous monitoring and real-time alerting mechanism, the system helps detect leaks early, reduce wastage of compressed air, minimize energy loss, and improve overall industrial safety. The working principle is simple, efficient, and suitable for both small and large industries.

WORK FLOW DIAGRAM :

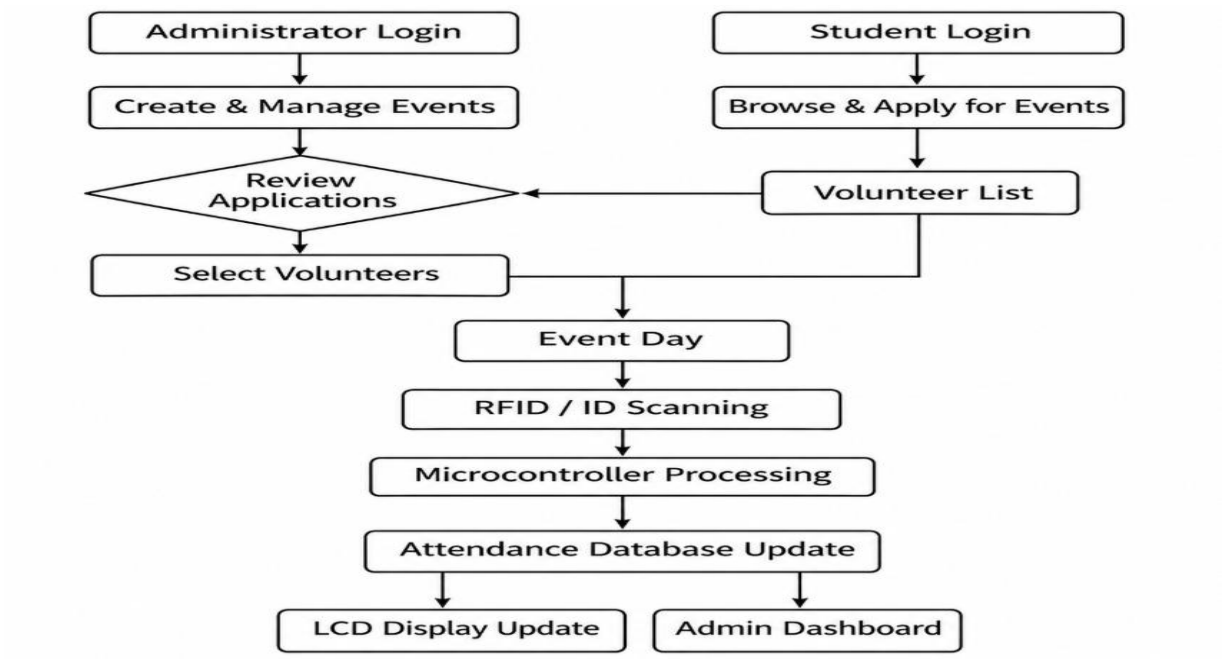

Figure 5.1 Flow Diagram

Advantages of the System

- Prevents financial losses caused by compressed air leakage
- Enhances operational safety by identifying potential pipeline failures
- Reduces manual labour and inspection time
- Improves machine performance and increases system life
- Low-power and economical solution suitable for small and large industries.

Proposed System Components:

Component	Use
ESP32	Processes RFID data and controls attendance system
RFID Reader	Scans student ID cards for attendance marking
RFID Tags	Stores unique identification data of each student.
LCD Display	Displays student ID and attendance status
Web Application	Manages events, applications, and attendance records
Database	Stores student, event, and attendance data locally
Wi-Fi Module	Enables communication between hardware and web system

Table 5.1 Proposed System Components

BLOCK DIAGRAM

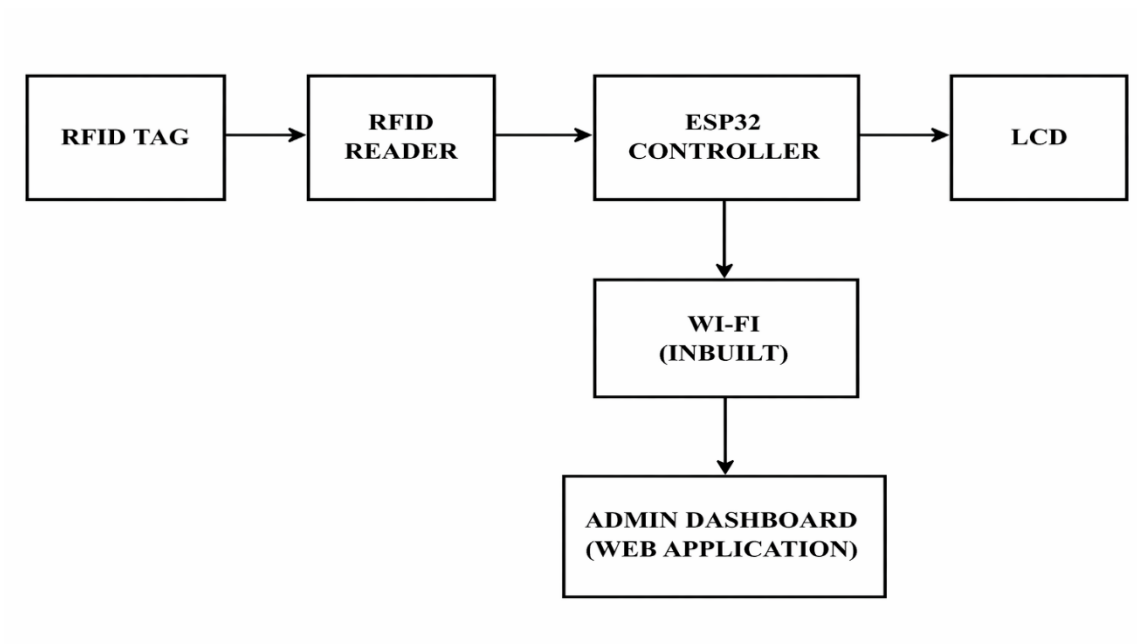

Figure 5.2 Block Diagram

FRONT-END (USER INTERFACE)

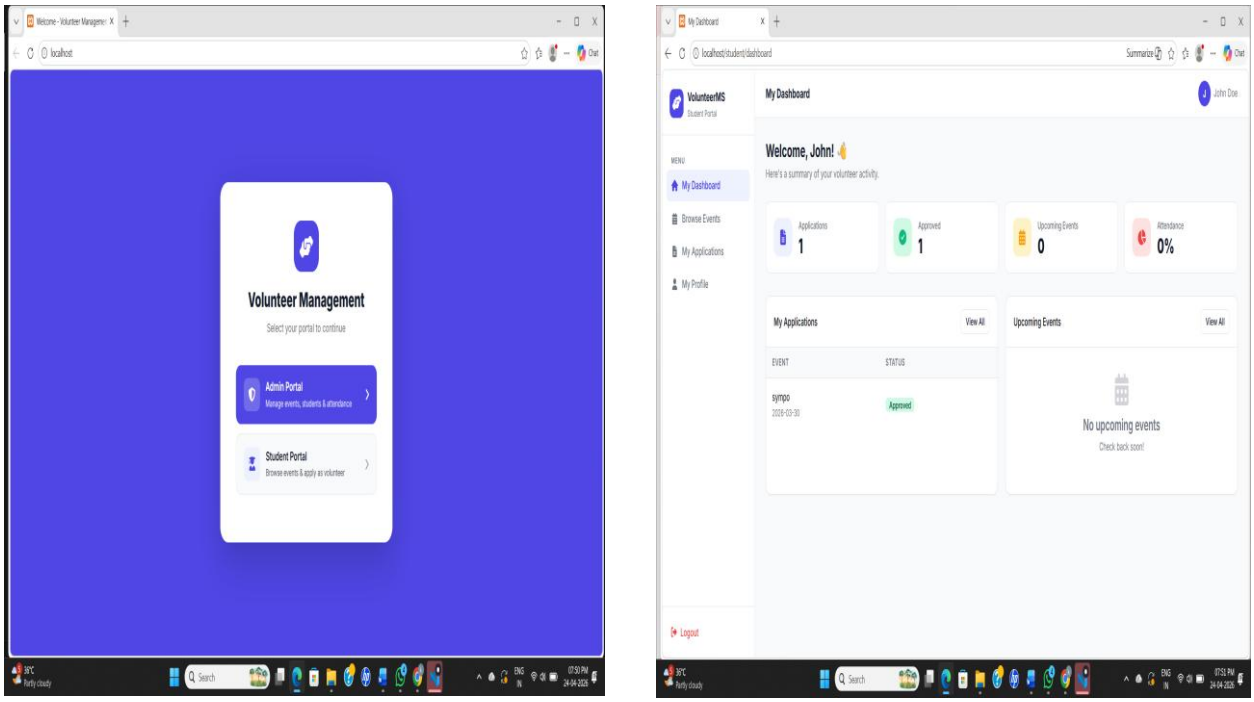

Figure 5.3 Front-End (User Interface)

BACK-END (SERVER SIDE)

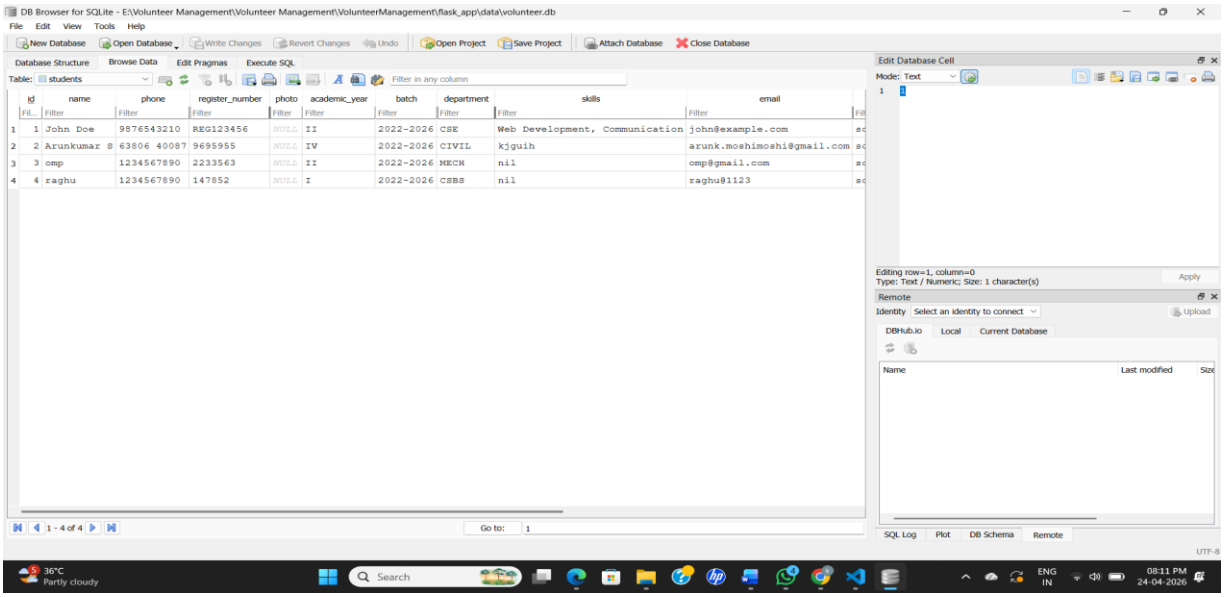

Figure 5.4 Back-End (Server Side)

HARDWARE SETUP

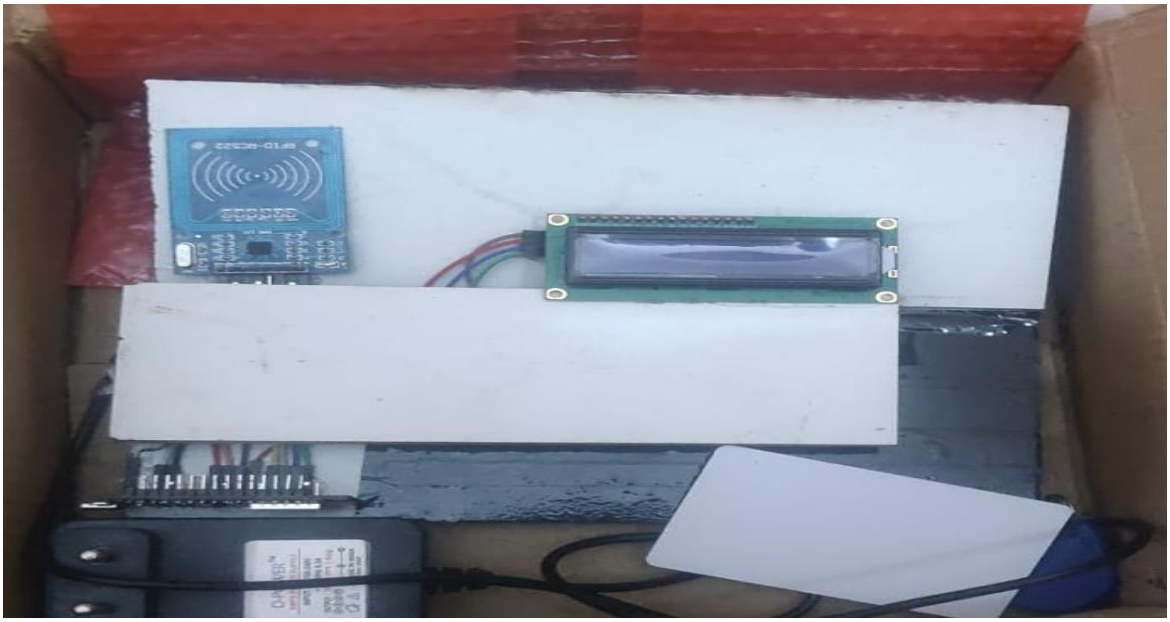

Figure 5.5 Hardware Setup

ATTENDANCE MARKING

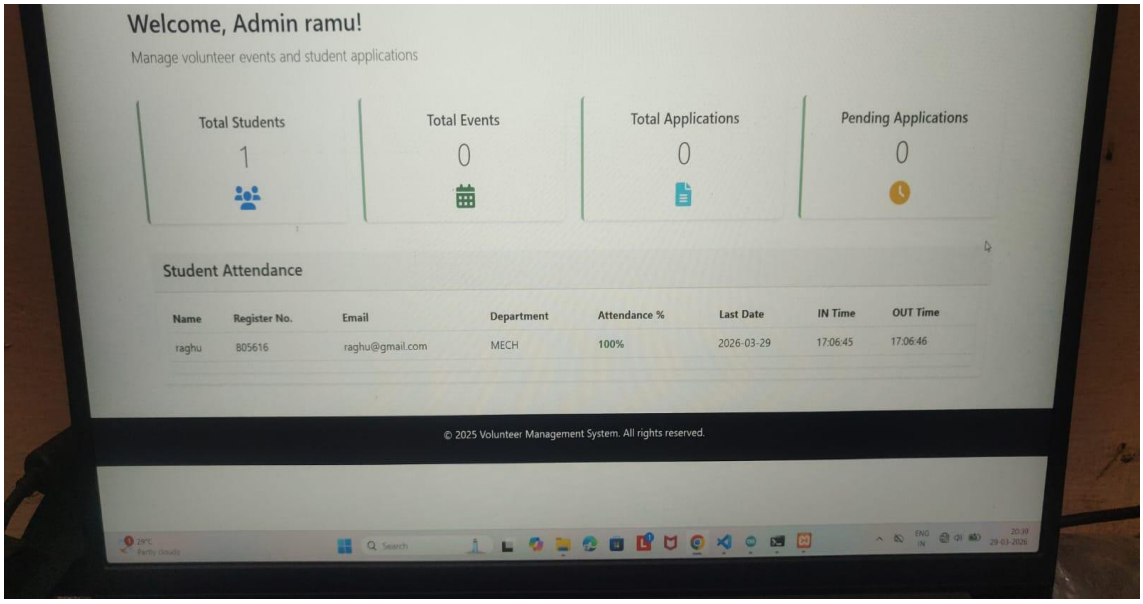

Figure 5.6 Attendance Marking

CHAPTER 6

HARDWARE AND SOFTWARE DESCRIPTION

6.1 Introduction

This chapter explains the software and hardware components used in the development of the Volunteer Management System for college events. The system is designed to manage volunteer registration, attendance tracking, and monitoring through an automated process. It uses hardware components for data collection and software technologies for data processing, storage, and display through an admin dashboard. The integration of both ensures accuracy, efficiency, and real-time updates.

6.2 Hardware Description

The hardware components are responsible for collecting input data and providing alerts or display outputs.

6.2.1 ESP32 Microcontroller

The ESP32 microcontroller acts as the central processing unit of the system. It reads input data from connected modules and sends it to the backend server. It has built-in Wi-Fi capability, which helps in transmitting data to the admin dashboard in real time. It is fast, efficient, and suitable for IoT-based applications.

6.2.2 RFID Module

The RFID module is used to identify volunteers. Each volunteer is provided with an RFID tag/card. When the card is scanned, the system reads the unique ID and sends it to the ESP32 for processing. This helps in automatic attendance marking.

6.2.3 LCD Display

The LCD display is used to show real-time information such as volunteer name, ID status, and confirmation messages like “Attendance Marked”. It provides immediate feedback to the user.

6.2.4 Power Supply

A regulated power supply is used to provide stable voltage to all the hardware components. Proper power supply ensures smooth and uninterrupted operation of the system.

6.3 Software Description

The software part of the Volunteer Management System plays a crucial role in processing data, storing information, and displaying it to the user through a web-based dashboard. It ensures smooth communication between the hardware (ESP32 & RFID) and the user interface.

6.3.1 Arduino IDE

Arduino IDE is used to develop and upload programs to the ESP32 microcontroller. It provides an easy environment to write, compile, and debug code. In this project, Arduino IDE is used to interface the RFID module, LCD display, and buzzer with ESP32. It reads RFID card data, processes it, and sends it to the backend server using Wi-Fi. The IDE also helps in serial monitoring for testing and debugging during development.

6.3.2 Embedded C

Embedded C programming is used to implement the logic inside the ESP32. It controls how the system behaves when a volunteer scans their RFID card. The program checks the input, sends data to the server, and activates output devices like LCD and buzzer. It also handles conditions such as valid card detection,

duplicate entry prevention, and error handling. This ensures the system works efficiently and accurately.

6.3.3 Flask Framework (Backend)

Flask is a lightweight Python-based web framework used to develop the backend of the system. It acts as a bridge between the hardware and the database. When ESP32 sends RFID data, Flask receives it through HTTP requests, processes it, and stores it in the database. Flask also handles routing, server-side logic, and communication with the frontend. It enables real-time updating of attendance data and ensures smooth data flow in the system.

6.3.4 HTML & CSS (Frontend Dashboard)

HTML and CSS are used to design the admin dashboard interface. HTML provides the structure of the webpage, while CSS is used for styling and layout. The dashboard displays volunteer details, attendance status, timestamps, and other relevant data in a clear format. It allows the admin to monitor event participation easily. The interface is designed to be simple, user-friendly, and responsive.

6.3.5 SQLite Database

SQLite is used as the database to store all system data. It maintains records such as volunteer ID, name, event details, and attendance logs. SQLite is lightweight, does not require a separate server, and is easy to integrate with Flask. It ensures quick data retrieval and efficient storage, making it suitable for small to medium-scale applications like this project.

6.3.6 System Integration

All software components work together to form a complete system. The ESP32 sends RFID data to the Flask server, which processes and stores it in SQLite. The

frontend dashboard retrieves this data and displays it to the admin in real time. This integration ensures automation, reduces manual errors, and improves overall system efficiency.

CIRCUIT DIAGRAM

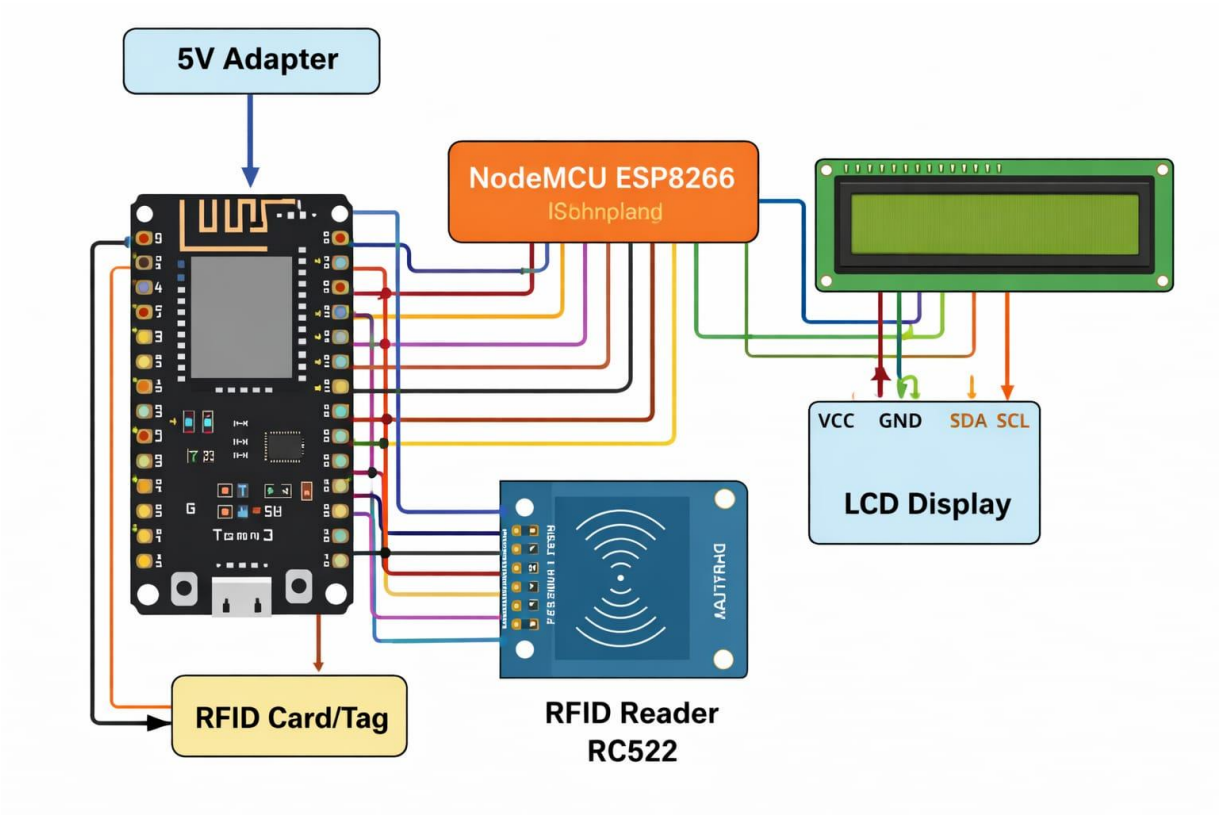

Figure 6.1 Circuit Diagram

CHAPTER 7

RESULTS AND DISCUSSION

RESULTS

The Volunteer Management System was successfully developed and tested to evaluate its performance in real-time conditions. During testing, each volunteer used an RFID card, and the system was able to read and process the card data quickly using the ESP32 microcontroller. The data was transmitted to the backend server through Wi-Fi with minimal delay, ensuring smooth communication between hardware and software.

The LCD display provided immediate feedback by showing messages such as "Attendance Marked," "Invalid Card," and "Already Recorded." The buzzer also generated an alert during each scan, improving user interaction and confirmation. This ensured that users clearly understood whether their attendance was recorded or not. On the software side, the Flask backend efficiently handled incoming data and stored it in the SQLite database. The stored data included important details such as volunteer ID and timestamp, and no duplication or data loss was observed during testing.

The admin dashboard displayed attendance records in real time, allowing organizers to monitor participation easily. Compared to manual methods, the system significantly reduced time consumption, minimized human errors, and improved accuracy.

Overall, the system performed reliably and achieved the objective of automating volunteer attendance management for college events.

DISCUSSION

The results demonstrate that the proposed system is reliable, efficient, and user-friendly. Compared to manual methods, the automated system provides real-time monitoring and better transparency in volunteer management. The use of RFID technology ensures quick identification, while the integration of ESP32 and Flask enables seamless communication between hardware and software.

One of the key advantages observed is the reduction in time required for attendance tracking. The system also minimizes human errors such as duplicate entries or missed records. The admin dashboard provides a centralized platform for monitoring, which improves decision-making during events.

However, certain limitations were identified. The system depends on a stable Wi-Fi connection for real-time data transmission. In case of network failure, there may be delays in updating the dashboard. Additionally, the system is currently designed for small to medium-scale use and may require further optimization for large-scale deployments.

Future improvements can include cloud database integration, mobile application support, and enhanced security features. Despite these limitations, the system performs effectively and meets the objectives of automating volunteer management in college events.

Attendance Update Status :

Action	Dashboard Output
RFID Card Scanned	Attendance Marked as Present
No Scan	No Update in Dashboard

Table 7.1 Attendance Update Status

Experimental Results :

Test no	RFID Scan	System condition	LCD Display	Dashboard Update
1	Card Scanned	Attendance Recorded	Attendance Marked	Yes
2	Card Scanned	Attendance Recorded	Attendance Marked	Yes
3	No Scan	Idle	Waiting for Card	No
4	Card Scanned	Attendance Recorded	Attendance Marked	Yes
5	No Scan	Idle	Waiting for Card	No

Table 7.2 Experimental Results

CHAPTER 8

CONCLUSION AND FUTURE WORK

CONCLUSION

The Volunteer Management System for college events provides an efficient, reliable, and low-cost solution for managing volunteer attendance using RFID technology. By utilizing an RFID module, ESP32 microcontroller, and a web-based dashboard, the system automates the process of attendance tracking in real time. This approach overcomes the limitations of traditional manual methods, which are time-consuming, error-prone, and difficult to manage during large events.

The implementation of real-time dashboard updates ensures that event organizers can instantly monitor volunteer participation without any delay. The system eliminates issues such as incorrect entries, missed attendance, and manual record maintenance. As a result, it reduces administrative workload, saves time, and improves overall efficiency in event management.

The integration of hardware and software components ensures smooth data communication, accurate data storage, and user-friendly interaction. The system proves that with simple and cost-effective components, an effective attendance management solution can be developed for practical use in educational institutions.

Overall, the proposed system serves as a scalable and efficient solution that enhances transparency, accuracy, and ease of managing volunteers during college events.

FUTURE WORK

The current system provides a strong foundation for automated volunteer attendance management; however, several improvements can be implemented to enhance its functionality and scalability. In future, the system can be expanded to support multiple events and a larger number of volunteers by integrating a cloud-based database for better data management and accessibility.

Advanced features such as mobile application integration can be added to allow organizers to monitor attendance from anywhere. Security can be improved by incorporating biometric or face recognition systems along with RFID to ensure more reliable authentication.

Real-time notifications can be implemented to inform organizers about attendance updates instantly. Additionally, the dashboard can be enhanced with data analytics features such as attendance reports, graphs, and performance tracking for better decision-making.

The system can also be integrated with existing college management systems to create a unified platform for student and event management. With these improvements, the project can evolve into a complete smart event management system aligned with modern digital and automation trends.

APPENDICES

```
// SOURCE CODE
@echo off
TITLE Volunteer Management System
color 0A
cls
echo.
echo  ==========================================================
echo      Volunteer Management System  ^|  Python Edition
echo  ==========================================================
echo.

:: ── Check Python is installed ──────────────────────────────
where python >nul 2>&1
if %errorlevel% neq 0 (
    color 0C
    echo  [ERROR] Python is NOT installed on this PC.
    echo.
    echo  Please install Python 3.10 or newer from:
    echo     https://www.python.org/downloads/
    echo.
    echo  IMPORTANT: During install, tick "Add Python to PATH"
    echo.
    pause
    exit /b 1
)
```

```
for /f "tokens=*" %%v in ('python --version 2^>^&1') do set PYVER=%%v
echo  [OK] Found %PYVER%
echo.

:: ── Install dependencies (only first time) ──────────────────
echo  Installing/checking required packages...
pip install flask werkzeug --quiet --disable-pip-version-check
if %errorlevel% neq 0 (
   color 0C
   echo  [ERROR] Failed to install packages. Check your internet connection.
   pause
   exit /b 1
)
echo  [OK] All packages ready.
echo.
:: ── Start the Flask app ──────────────────────────────────────
echo  Starting server on http://localhost ...
echo.
echo  *** LEAVE THIS WINDOW OPEN ***
echo  To stop the server press Ctrl+C or close this window.
echo  ═══════════════════════════════════════════════════════
echo.
:: Wait 2 seconds then open browser
ping -n 3 127.0.0.1 >nul
start http://localhost

:: Run Flask app
```

```
python "%~dp0flask_app\app.py"

echo.
echo  Server stopped.
Pause
//FRONT END
% extends "admin_base.html" %}
{% block title %}Applications{% endblock %}
{% block page_title %}Applications{% endblock %}
{% block content %}
<div class="page-header"><h1>Applications</h1><p>Review and approve volunteer applications</p></div>
<div class="card">
  <div style="overflow-x:auto;">
    <table class="table">
      <thead><tr>
        <th>Student</th><th>Event</th><th>Event Date</th>
        <th>Applied On</th><th>Attendance</th><th>Status</th><th>Action</th>
      </tr></thead>
      <tbody>
      {% for app, att_pct in apps %}
        <tr>
          <td>
            <div class="table-name">{{ app['student_name'] }}</div>
             <div style="font-size:0.78rem;color:#6B7280;">{{ app['register_number'] }}</div>
          </td>
```

```
<td>{{ app['event_title'] }}</td>
<td style="font-size:0.82rem;">{{ app['event_date'] }}</td>
<td style="font-size:0.82rem;color:#6B7280;">{{ app['applied_at'][:10] }}</td>
<td>
<span style="font-size:0.82rem;font-weight:600;color:{{ '#10B981' if att_pct >= 75 else '#EF4444' }};">
{{ att_pct }}%
</span>
</td>
<td>
<span class="badge badge-{{ app['status']|lower }}">{{ app['status'] }}</span>
</td>
<td>
<form method="POST" style="display:flex;gap:6px;align-items:center;">
<input type="hidden" name="application_id" value="{{ app['id'] }}">
<select name="status" class="form-select" style="width:110px;padding:5px 8px;font-size:0.8rem;">
<option {% if app['status']=='Pending' %}selected{% endif %}>Pending</option>
<option {% if app['status']=='Approved' %}selected{% endif %}>Approved</option>
<option {% if app['status']=='Rejected' %}selected{% endif %}>Rejected</option>
</select>
<button class="btn btn-primary btn-sm">Save</button>
```

```
            </form>
          </td>
        </tr>
      {% else %}
        <tr><td colspan="7">
          <div class="empty-state"><i class="fas fa-file-alt"></i>
            <h5>No applications yet</h5>
            <p>Students will appear here after applying to events.</p>
          </div>
        </td></tr>
      {% endfor %}
      </tbody>
    </table>
  </div>
</div>
{% endblock %}
{% extends "base.html" %}
{% block body %}
<div class="overlay" id="overlay" onclick="toggleSidebar()"></div>

<!-- Sidebar -->
<aside class="sidebar" id="sidebar">
  <div class="sidebar-logo">
    <div class="logo-icon"><i class="fas fa-hands-helping"></i></div>
    <div>
      <span>VolunteerMS</span>
      <small>Admin Panel</small>
```

```
    </div>
  </div>
  <nav>
    <div class="sidebar-section">Main</div>
        <a href="/admin/dashboard" class="{{ 'active' if request.endpoint ==
'admin_dashboard' }}">
      <i class="fas fa-home"></i> Dashboard
    </a>
    <a href="/admin/events" class="{{ 'active' if 'event' in request.endpoint }}">
      <i class="fas fa-calendar-alt"></i> Events
    </a>
    <a href="/admin/students" class="{{ 'active' if 'student' in request.endpoint }}">
      <i class="fas fa-user-graduate"></i> Students
    </a>
        <a href="/admin/applications" class="{{ 'active' if 'application' in
request.endpoint }}">
      <i class="fas fa-file-alt"></i> Applications
    </a>
    <div class="sidebar-section">Setup</div>
     <a href="/admin/hardware" class="{{ 'active' if 'hardware' in request.endpoint
}}">
      <i class="fas fa-microchip"></i> Hardware Setup
    </a>
    <a href="/admin/register">
      <i class="fas fa-user-plus"></i> Add Admin
    </a>
  </nav>
```

```
  <div class="sidebar-bottom">
    <a href="/admin/logout"><i class="fas fa-sign-out-alt"></i> Logout</a>
  </div>
</aside>

<!-- Main -->
<div class="main-wrap">
  <div class="topbar">
    <div style="display:flex;align-items:center;gap:12px;">
        <button class="hamburger" onclick="toggleSidebar()"><i class="fas fa-
bars"></i></button>
      <span class="topbar-title">{% block page_title %}Dashboard{% endblock
%}</span>
    </div>
    <div class="topbar-user">
        <div class="user-avatar">{{ session.get('admin_username', 'A')[0]|upper
}}</div>
      <span>{{ session.get('admin_username', 'Admin') }}</span>
    </div>
  </div>

  <div class="page-content">
    {% with messages = get_flashed_messages(with_categories=true) %}
      {% for cat, msg in messages %}
        <div class="alert alert-{{ 'success' if cat == 'success' else 'danger' }}">
          <i class="fas fa-{{ 'check-circle' if cat == 'success' else 'exclamation-circle'
}}"></i>
```

```
        {{ msg }}
      </div>
     {% endfor %}
    {% endwith %}
    {% block content %}{% endblock %}
  </div>
</div>
{% endblock %}
// DATABASE
import sqlite3
import os

DB_PATH = os.path.join(os.path.dirname(__file__), 'data', 'volunteer.db')

def get_db():
    """Get a database connection."""
    os.makedirs(os.path.dirname(DB_PATH), exist_ok=True)
    conn = sqlite3.connect(DB_PATH)
    conn.row_factory = sqlite3.Row  # returns dict-like rows
    conn.execute("PRAGMA journal_mode=WAL")
    return conn

def init_db():
    """Create all tables and default data if they don't exist."""
    conn = get_db()
    c = conn.cursor()
```

```
c.executescript("""
    CREATE TABLE IF NOT EXISTS admins (
        id       INTEGER PRIMARY KEY AUTOINCREMENT,
        username TEXT UNIQUE NOT NULL,
        email    TEXT UNIQUE NOT NULL,
        password TEXT NOT NULL,
        created_at TEXT DEFAULT (datetime('now'))
    );

    CREATE TABLE IF NOT EXISTS students (
        id              INTEGER PRIMARY KEY AUTOINCREMENT,
        name            TEXT NOT NULL,
        phone           TEXT NOT NULL,
        register_number TEXT UNIQUE NOT NULL,
        photo           TEXT,
        academic_year   TEXT NOT NULL,
        batch           TEXT NOT NULL,
        department      TEXT NOT NULL,
        skills          TEXT,
        email           TEXT UNIQUE NOT NULL,
        password        TEXT NOT NULL,
        rfid_tag        TEXT DEFAULT NULL,
        event_id        INTEGER DEFAULT NULL,
        created_at      TEXT DEFAULT (datetime('now'))
    );

    CREATE TABLE IF NOT EXISTS events (
```

```
    id              INTEGER PRIMARY KEY AUTOINCREMENT,
    title           TEXT NOT NULL,
    description     TEXT NOT NULL,
    brochure        TEXT,
    date            TEXT NOT NULL,
    time            TEXT NOT NULL,
    venue           TEXT NOT NULL,
    required_skills TEXT,
    priority        TEXT DEFAULT 'Medium',
    max_volunteers  INTEGER NOT NULL,
    event_code      TEXT DEFAULT NULL,
    admin_id        INTEGER,
    created_at      TEXT DEFAULT (datetime('now')),
    FOREIGN KEY (admin_id) REFERENCES admins(id)
);

CREATE TABLE IF NOT EXISTS applications (
    id         INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER,
    event_id   INTEGER,
    status     TEXT DEFAULT 'Pending',
    applied_at TEXT DEFAULT (datetime('now')),
    FOREIGN KEY (student_id) REFERENCES students(id),
    FOREIGN KEY (event_id)   REFERENCES events(id)
);

CREATE TABLE IF NOT EXISTS attendance (
```

```
        id              INTEGER PRIMARY KEY AUTOINCREMENT,
        rfid_tag        TEXT NOT NULL,
        name            TEXT,
        register_number TEXT,
        date            TEXT NOT NULL,
        in_time         TEXT DEFAULT NULL,
        out_time        TEXT DEFAULT NULL,
        event_code      TEXT DEFAULT NULL,
        created_at      TEXT DEFAULT (datetime('now'))
    );
""")

# Insert default admin if not exists (password: admin123)
from werkzeug.security import generate_password_hash
c.execute("SELECT id FROM admins WHERE username='admin'")
if not c.fetchone():
    c.execute(
        "INSERT INTO admins (username, email, password) VALUES (?,?,?)",
        ('admin', 'admin@example.com', generate_password_hash('admin123'))
    )

conn.commit()
conn.close()
print("[DB] Database initialized at:", DB_PATH)
// HARDWARE
#include <WiFi.h>
#include <HTTPClient.h>
```

```
#include <SPI.h>
#include <MFRC522.h>
#include <LiquidCrystal_I2C.h>

// WiFi Credentials
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// Server URL
const char* serverName = "http://YOUR_SERVER_IP:5000/attendance";

// RFID Pins
#define SS_PIN 5
#define RST_PIN 22
MFRC522 rfid(SS_PIN, RST_PIN);

// LCD (I2C Address 0x27, 16x2)
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  Serial.begin(115200);

  // Initialize SPI & RFID
  SPI.begin();
  rfid.PCD_Init();

  // Initialize LCD
```

```
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Scan Your Card");

  // Connect WiFi
  WiFi.begin(ssid, password);
  Serial.print("Connecting");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nConnected to WiFi");
}

void loop() {
  // Check for RFID card
  if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) {
    return;
  }

  // Read UID
  String cardID = "";
  for (byte i = 0; i < rfid.uid.size; i++) {
    cardID += String(rfid.uid.uidByte[i], HEX);
  }
```

```
Serial.println("Card ID: " + cardID);

// Display on LCD
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Processing...");

// Send to Server
if (WiFi.status() == WL_CONNECTED) {
  HTTPClient http;
  http.begin(serverName);
  http.addHeader("Content-Type", "application/json");

  String jsonData = "{\"rfid\":\"" + cardID + "\"}";
  int httpResponseCode = http.POST(jsonData);

  if (httpResponseCode > 0) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Attendance Done");
    Serial.println("Data Sent Successfully");
  } else {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Error Sending");
    Serial.println("Error in sending data");
  }
```

```
    http.end();
  }

  delay(3000);

  // Reset LCD
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Scan Your Card");

  rfid.PICC_HaltA();
}
```

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